

PRACTICE SHEET-1

1. If the height of a given cone be doubled and radius of the base remains the same, the ratio of the volume of the given cone to that of the second cone will be
 (1) 2 : 1 (2) 1 : 8
 (3) 1 : 2 (4) 8 : 1
2. If the radius of the base of a cone be doubled and height is left unchanged, then ratio of the volume of new cone to that of the original cone will be:
 (1) 1 : 4 (2) 2 : 1
 (3) 1 : 2 (4) 4 : 1
3. Each of the measure of the radius of base of a cone and that of a sphere is 8 cm. Also, the volume of these two solids are equal. The slant height of the cone is
 (1) $8\sqrt{17}$ cm (2) $4\sqrt{17}$ cm
 (3) $34\sqrt{2}$ cm (4) 34 cm.
4. A cone of height 15 cm and base diameter 30 cm is carved out of a wooden sphere of radius 15 cm. The percentage of wasted wood is:
 (1) 75% (2) 50%
 (3) 40% (4) 25%
5. In a right circular cone, the radius of its base is 7 cm and its height 24 cm. A cross-section is made through the midpoint of the height parallel to the base. The volume of the upper portion is
 (1) 169 cm^3 (2) 154 cm^3
 (3) 1078 cm^3 (4) 800 cm^3
6. If the area of the base of a cone is 770 cm^2 and the area of the curved surface is 814 cm^2 , then its volume (in cm^3) is:
 (1) $213\sqrt{5}$ (2) $392\sqrt{5}$
 (3) $550\sqrt{5}$ (4) $616\sqrt{5}$
7. Volume of two cones are in the ratio 1 : 4 and their diameters are in the ratio 4 : 5. The ratio of their height is
 (1) 1 : 5 (2) 5 : 4
 (3) 5 : 16 (4) 25 : 64
8. The height of the cone is 30 cm. A small cone is cut off at the top by a plane parallel to its base. If its volume is $\frac{1}{27}$ of the volume of the cone, at what height, above the base, is the section made?
 (1) 6 cm (2) 8 cm
 (3) 10 cm (4) 20 cm
9. The radius of the base and height of a right circular cone are in the ratio 5 : 12. If the volume of the cone is $314\frac{2}{7}\text{ cm}^3$, the slant height (in cm) of the cone will be
 (1) 12 (2) 13
 (3) 15 (4) 17
10. Two solid right cones of equal height and of radii r_1 and r_2 are melted and made to form a solid sphere of radius R. Then the height of the cone is
 (1) $\frac{4R^2}{r_1^2 + r_2^2}$ (2) $\frac{4R}{r_1 + r_2}$
 (3) $\frac{4R^3}{r_1^2 + r_2^2}$ (4) $\frac{R^2}{r_1^2 + r_2^2}$
11. The ratio of radii of two cone is 3 : 4 and the ratio of their height is 4 : 3. Then the ratio of their volume will be
 (1) 3 : 4 (2) 4 : 3
 (3) 9 : 16 (4) 16 : 9
12. If a right circular cone is separated into solids of volumes V_1, V_2, V_3 by two planes parallel to the base, which also trisect the altitude, then $V_1 : V_2 : V_3$ is
 (1) 1 : 2 : 3 (2) 1 : 4 : 6
 (3) 1 : 6 : 9 (4) 1 : 7 : 19
13. If the radii of the circular ends of a truncated conical bucket which is 45cm high be 28 cm and 7 cm, then the capacity of the bucket in cubic centimetre is
 (1) 48510 (2) 45810
 (3) 48150 (4) 48051
14. The ratio of height and the diameter of a right circular cone is 3 : 2 and its volume is 1078 cc , then its height is :
 (1) 7 cm (2) 14 cm
 (3) 21 cm (4) 28 cm

15. The radius of the base of a right circular cone is doubled keeping its height fixed. The volume of the cone will be:
 (1) Three times of the previous volume
 (2) Four times of the previous volume
 (3) 2 times of the previous volume
 (4) Double of the previous volume
16. The heights of two cones are in the ratio 1 : 3 and the diameters of their base are in the ratio 3 : 5. The ratio of their volume is
 (1) 3 : 25 (2) 4 : 25
 (3) 6 : 25 (4) 7 : 25
17. The base of a right circular cone has the same radius a as that of a sphere. Both the sphere and the cone have the same volume. Height of the cone is
 (1) $3a$ (2) $4a$
 (3) $\frac{7}{4}a$ (4) $\frac{7}{3}a$
18. The circumference of the base of a 16 cm height solid cone is 33 cm. What is the volume of the cone in cm^3 ?
 (1) 1028 (2) 616
 (3) 462 (4) 828
19. The perimeter of the base of a right circular cone is 8 cm. If the height of the cone is 21 cm, then its volume is:
 (1) $108\pi \text{ cm}^3$ (2) $112\pi \text{ cm}^3$
 (3) $112\pi \text{ cm}^3$ (4) $108\pi \text{ cm}^3$
20. If the volume of two right circular cones are in the ratio 4 : 1 and their diameter are in the ratio 5 : 4, then the ratio of their height is
 (1) 25 : 16 (2) 25 : 64
 (3) 64 : 25 (4) 16 : 25
21. The volume of a conical tent is 1232 cu. m and the area of its base is 154 sq. m. Find the length of the canvas required to build the tent, if the canvas is 2 m in width.
 (1) 270 m (2) 272 m
 (3) 276 m (4) 275 m
22. If the ratio of the diameters of two right circular cones of equal height be 3 : 4, then the ratio of their volume will be
 (1) 3 : 4 (2) 9 : 16
 (3) 16 : 9 (4) 27 : 64
23. A hollow spherical metallic ball has an external diameter 6 cm and is $\frac{1}{2}$ cm thick. The volume of the ball (in cm^3) is
 (1) $41\frac{2}{3}$ (2) $37\frac{2}{3}$
 (3) $47\frac{2}{3}$ (4) $40\frac{2}{3}$
24. The sum of radii of two spheres is 10 cm and the sum of their volume is 880 cm^3 . What will be the product of their radii?
 (1) 21 (2) $26\frac{1}{3}$
 (3) $33\frac{1}{3}$ (4) 70
25. If the radius of a sphere is doubled, its volume becomes
 (1) double (2) four times
 (3) six times (4) eight times
26. The radii of two spheres are in the ratio 3 : 2. Their volume will be in the ratio :
 (1) 9 : 4 (2) 3 : 2
 (3) 8 : 27 (4) 27 : 8
27. The total surface area of a solid hemisphere is $108\pi \text{ cm}^2$. The volume of the hemisphere is
 (1) $72\pi \text{ cm}^3$ (2) $144\pi \text{ cm}^3$
 (3) $108\sqrt{6} \text{ cm}^3$ (4) $54\sqrt{6} \text{ cm}^3$
28. The largest sphere is carved out of a cube of side 7 cm. The volume of the sphere (in cm^3) will be
 (1) 718.66 (2) 543.72
 (3) 481.34 (4) 179.67
29. The surface areas of two spheres are in the ratio 4 : 9. Their volumes will be in the ratio
 (1) 2 : 3 (2) 4 : 9
 (3) 8 : 27 (4) 64 : 729
30. A sphere and a hemisphere have the same volume. The ratio of their radii is
 (1) 1 : 2 (2) 1 : 8
 (3) $1 : \sqrt{2}$ (4) $1 : \sqrt[3]{2}$

ANSWER KEY

1.	(3)	2.	(4)	3.	(1)	4.	(1)	5.	(2)	6.	(4)	7.	(4)	8.	(4)	9.	(2)	10.	(3)
11.	(1)	12.	(4)	13.	(1)	14.	(3)	15.	(2)	16.	(1)	17.	(2)	18.	(3)	19.	(2)	20.	(3)
21.	(4)	22.	(2)	23.	(3)	24.	(2)	25.	(4)	26.	(4)	27.	(2)	28.	(4)	29.	(3)	30.	(4)

PRACTICE SHEET-2

- A solid sphere of 6 cm diameter is melted and recast into 8 solid spheres of equal volume. The radius (in cm) of each small sphere is
 (1) 1.5 (2) 3
 (3) 2 (4) 2.5
- The total surface area of a sphere is 8π square unit. The volume of the sphere is
 (1) $\frac{8\sqrt{2}}{3}\pi$ cubic unit
 (2) $\frac{8}{3}\pi$ cubic unit
 (3) $8\sqrt{3}\pi$ cubic unit
 (4) $\frac{8\sqrt{3}}{5}\pi$ cubic unit
- Area of the base of a pyramid is 57 sq.cm. and height is 10 cm, then its volume (in cm^3), is
 (1) 570 (2) 390
 (3) 190 (4) 590
- There is a pyramid on a base which is a regular hexagon of side 2a cm. If every slant edge of this pyramid is of length $\frac{5a}{2}$ cm, then the volume of this pyramid is
 (1) $3a^3 \text{ cm}^3$ (2) $3\sqrt{2} a^3 \text{ cm}^3$
 (3) $3\sqrt{3} a^3 \text{ cm}^3$ (4) $6a^3 \text{ cm}^3$
- The base of a right pyramid is a square of side 40 cm long. If the volume of the pyramid is 8000 cm^3 , then its height is:
 (1) 5 cm (2) 10 cm
 (3) 15 cm (4) 20 cm
- The base of a right prism is a trapezium. The length of the parallel sides are 8 cm and 14 cm and the distance between the parallel sides is 8 cm. If the volume of the prism is 1056 cm^3 , then the height of the prism is
 (1) 44 cm (2) 16.5 cm
 (3) 12 cm (4) 10.56 cm
- The height of a right prism with a square base is 15 cm. If the area of the total surface of the prism is 608 sq. cm, its volume is
 (1) 910 cm^3 (2) 920 cm^3
 (3) 960 cm^3 (4) 980 cm^3
- The base of a right prism is an equilateral triangle of side 8 cm and height of the prism is 10 cm. Then the volume of the prism is
 (1) $320\sqrt{3}$ cubic cm (2) $160\sqrt{3}$ cubic cm
 (3) $150\sqrt{3}$ cubic cm (4) $300\sqrt{3}$ cubic cm
- The base of right prism is a triangle whose perimeter is 28 cm and the in radius of the triangle is 4 cm. If the volume of the prism is 366 cc, then its height is
 (1) 6 cm (2) 8 cm
 (3) 4 cm (4) None of these
- If the base of a right pyramid is triangle of sides 5 cm, 12 cm, 13 cm and its volume is 330 cm^3 , then its height (in cm) will be
 (1) 33 (2) 32
 (3) 11 (4) 22
- The diameter of the moon is assumed to be one fourth of the diameter of the earth. Then the ratio of the volume of the earth to that of the moon is
 (1) 64 : 1 (2) 1 : 64
 (3) 60 : 7 (4) 7 : 60
- A conical vessel whose internal radius is 12 cm and height 50 cm is full of liquid. The contents are emptied into a cylindrical vessel with radius (internal) 10 cm. The height to

which the liquid rises in the cylindrical vessel is:

- (1) 25cm (2) 20cm
(3) 24cm (4) 22cm

13. The volume of a right circular cylinder is equal to the volume of that right circular cone whose height is 108 cm and diameter of base is 30 cm. If the height of the cylinder is 9 cm, the diameter of its base is

- (1) 30 cm (2) 60 cm
(3) 50 cm (4) 40 cm

14. The total surface area of a cube and a sphere are equal. What will be the ratio between their volume?

- (1) $\pi : 6$ (2) $\sqrt{\pi} : \sqrt{6}$
(3) $\sqrt{6} : \sqrt{\pi}$ (4) $6 : \pi$

15. A rectangular paper sheet of dimensions 22 cm $\times$ 12 cm is folded in the form of a cylinder along its length. What will be the volume of this cylinder?

- (1) 460 cm³ (2) 462 cm³
(3) 624 cm³ (4) 400 cm³

16. The ratio of the volume of a cube to that of a sphere, which will fit exactly inside the cube, is

- (1) $\pi : 6$ (2) $6 : \pi$
(3) $6 : \pi$ (4) $\pi : 3$

17. The volume of a sphere and a right circular cylinder having the same radius are equal, The ratio of the diameter of the sphere to the height of the cylinder is

- (1) 3 : 2 (2) 2 : 3
(3) 1 : 2 (4) 2 : 1

18. The size of a rectangular piece of paper is 100 cm $\times$ 44 cm. A cylinder is formed by rolling the paper along its length. The volume of the cylinder is

- (1) 4400 cm³ (2) 15400 cm³
(3) 35000 cm³ (4) 144 cm³

19. A cone, a hemisphere and a cylinder stand on equal bases and have the same height. The ratio of their respective volume is

- (1) 1 : 2 : 3 (2) 2 : 1 : 3
(3) 1 : 3 : 2 (4) 3 : 1 : 2

20. The height of a cylinder and that of a cone are in the ratio 2 : 3 and the radii of their bases in the ratio 3 : 4. The ratio of their volume will be

- (1) 1 : 9 (2) 2 : 9
(3) 9 : 8 (4) 3 : 8

21. Find the length of the largest rod that can be placed in a room 16m long, 12m broad and $10\frac{2}{3}$ m. high.

- (1) 23 m. (2) 68 m.
(3) $22\frac{2}{3}$ m. (4) $22\frac{1}{3}$ m.

22. If the volume of two cubes are in the ratio 27 : 64, then the ratio of their total surface area is :

- (1) 27 : 64 (2) 3 : 4
(3) 9 : 16 (4) 3 : 8

23. Find the length of the longest rod that can be placed in a hall of 10 m length, 6 m breadth and 4 m height.

- (1) $2\sqrt{38}$ m (2) $4\sqrt{38}$ m
(3) $2\sqrt{19}$ m (4) 19 m

24. The volume of a cuboid is twice the volume of a cube. If the dimensions of the cuboid are 9 cm, 8 cm and 6 cm, the total surface area of the cube is :

- (1) 72 cm² (2) 216 cm²
(3) 432 cm² (4) 108 cm²

25. The length, breadth and height of a room is 5m, 4m and 3m respectively. Find the length of the largest bamboo that can be kept inside the room.

- (1) 5 m (2) 60 m
(3) 7 m (4) $5\sqrt{2}$ m

26. The length of the longest rod that can be placed in a room which is 12 m long, 9 m broad and 8 m high is

- (1) 27 m (2) 19 m
(3) 17 m (4) 13 m

27. A cube of edge 5 cm is cut into cubes each of edge of 1 cm. The ratio of the total surface area of one of the small cubes to that of the large cube is equal to :

- (1) 1 : 125 (2) 1 : 5
(3) 1 : 625 (4) 1 : 25

28. The perimeter of the floor of a room is 18 m. What is the area of the walls of the room, if the height of the room is 3 m ?

- (1) 21 m² (2) 42 m²
(3) 54 m² (4) 108 m²

29. The length (in metres) of the longest rod that can be put in a room of dimensions $10\text{ m} \times 10\text{ m} \times 5\text{ m}$ is
 (1) $15\sqrt{3}$ (2) 15
 (3) $10\sqrt{2}$ (4) $5\sqrt{3}$
30. The floor of a room is of size $4\text{ m} \times 3\text{ m}$ and its height is 3 m. The walls and ceiling of the room require painting. The area to be painted is
 (1) 66 m^2 (2) 54 m^2
 (3) 43 m^2 (4) 33 m^2

ANSWER KEY

1.	(1)	2.	(1)	3.	(3)	4.	(3)	5.	(3)	6.	(3)	7.	(4)	8.	(3)	9.	(4)	10.	(1)
11.	(1)	12.	(3)	13.	(2)	14.	(2)	15.	(2)	16.	(2)	17.	(1)	18.	(3)	19.	(1)	20.	(3)
21.	(3)	22.	(3)	23.	(1)	24.	(2)	25.	(4)	26.	(3)	27.	(4)	28.	(3)	29.	(2)	30.	(2)

CDS PYQ

1. What is the quantity of cloth required to roll up to form a right circular tent whose base is of radius 12m and height 5m?
[CDS-2013-1]
(a) 40π sq m (b) 60π sq m
(c) 78π sq m (d) 156π
2. The length, breadth and height of a box are respectively 14m, 12m and 13m. The length of the greatest rod that can be put in it is:
[CDS-2013-1]
(a) 22.32m (b) 22.56m
(c) 20m (d) 19.5m
3. The volume of the material of a hemispherical shell with outer and inner radii 9cm and 7 cm, respectively is approximately:
[CDS-2013-1]
(a) 808cu cm (b) 800cu cm
(c) 816 cu cm (d) 824 cu cm
4. The ratio of surface area to diameter of a sphere whose volume is 36π cu cm is:
[CDS-2013-1]
(a) 3π (b) 6π
(c) 6 (d) None of these
5. A cylindrical tube open at both ends is made of metal. The internal diameter of the tube is 6cm and length of the tube is 10cm. If the thickness of the metal used is 1 cm, then the outer curved surface area of the tube is:
[CDS-2013-1]
(a) 140π sq cm (b) 146.5π sq. cm
(c) 70π sq cm (d) None of these
6. The volume of a right circular cone of height 3 cm and slant height 5 cm is:
[CDS-2013-1]
(a) 49.3cu cm (b) 50.3 cu cm
(c) 52 cu cm (d) 53 cu cm
7. A bucket is of a height 25cm. Its top and bottom radii are 20 cm and 10cm, respectively. Its capacity (in L) is:
[CDS-2013-1]
(a) $17.5\pi/3$ (b) 17.5π
(c) 20π (d) 25π
8. The height of a cylinder is 15cm. The lateral surface area is 660 sq. cm. Its volume is:
[CDS-2013-1]
(a) 1155 cu cm (b) 1215cu cm
(c) 1230 cu cm (d) 2310 cu cm
9. From a solid wooden right circular cylinder, a right circular cone whose radius and height are same as the radius and height of the cylinder, respectively is carved out. What is the ratio of the volume of the utilized wood to that of the wasted wood?
[CDS-2013-1]
(a) 1:2 (b) 2:1
(c) 2:3 (d) 1:3
10. A solid spherical ball of iron of radius 4cm is melted to form spheres of radius 2cm each. The number of spheres, so formed is:
[CDS-2013-1]
(a) 8 (b) 9
(c) 10 (d) 16
11. If the heights and the area of the base of a right circular cone and a pyramid with square base are the same, then they have:
[CDS-2013-1]
(a) Same volume and same surface area
(b) Same surface area but different volumes
(c) Same volume but different surface area
(d) Different volumes and different surface area
12. A swimming pool 70 meter long 44 meter wide and 3 m deep is field by water flowing from a pipe at the rate of 30800cm^3 per second the time taken to fill the swimming pool is:
[CDS-2013-1]
(a) $\frac{71}{2}$ hour (b) 80 hours
(c) $\frac{250}{3}$ hours (d) None of these
13. The diameter of the Moon is approximately one-fourth of the diameter of the Earth. What is the ratio (approximate) of their volumes?
[CDS-2013-2]
(a) 1:16 (b) 1:64
(c) 1:4 (d) 1:128
14. If the total surface area of a cube is 6 sq units, then what is the volume of the cube?

[CDS-2013-2]

- (a) 1 cu unit (b) 2 cu units
(c) 4 cu units (d) 6 cu units

15. Let A be a pyramid on a square base and B be a cube. If a, b and c denote the number of edges, number of faces and number of corners, respectively. They, the result $a = b + c$ is true for:

[CDS-2013-2]

- (a) Only A (b) Only B
(c) Both A and B (d) Neither A nor B

16. What is the volume of a cone having a base of radius 10cm and height 21 cm?

[CDS-2013-2]

- (a) 2200cm³ (b) 3000cm³
(c) 5600cm³ (d) 6600cm³

17. A conical cap has base diameter 24cm and height 16cm. What is the cost of painting the surface of the cap at the rate of 70 paise per sq cm?

[CDS-2013-2]

- (a) Rs520 (b) Rs524
(c) Rs528 (d) Rs532

18. What is the whole surface area of a cone of base radius 7 cm and height 24cm?

[CDS-2013-2]

- (a) 654sq cm (b) 704 sq cm
(c) 724 sq cm (d) 964 sq cm

19. A tent is in the form of a right circular cylinder surmounted by a cone. The diameter of the cylinder is 24m. The height of the cylindrical portion is 11m, while the vertex of the cone is 16m above the ground. What is the area of the curved surface for conical portion?

[CDS-2013-2]

- (a) 3434/9 sq m (b) 3431/8 sq cm
(c) 3432/7 sq m (d) 3234/7 sq cm

20. If x is the curved surface area and y is the volume of a right circular cylinder, then which one of the following is correct?

[CDS-2013-2]

- (a) Only the ratio of the height to radius of the cylinder is independent of x
(b) Only the ratio of height to radius of the cylinder is independent of y
(c) Either (a) or (b)
(d) Neither (a) nor (b)

21. A rectangular field is 22m long and 10m wide. Two hemispherical pitholes of radius 2m are dug from two places and the mud is spread over the remaining part of the field. The rise in the level of the field is:

[CDS 2013-2]

- (a) $\frac{8}{93}m$ (b) $\frac{16}{93}m$
(c) $\frac{13}{93}m$ (d) $\frac{23}{93}m$

22. A hospitals room is to accommodate 56 patients. It should be done in such a way that every patient gets 2.2 m² of floor and 8.8m² of space. If the length of the room is 14m, then breadth and the height of the room are respectively:

[CDS 2013-2]

- (a) 8.8m, 4m (b) 8.4m, 4.2m
(c) 8m, 4m (d) 7.8m, 4.2m

23. A cylinder is surmounted by a cone at one end, a hemisphere at the other end. The common radius is 3.5 cm, the height of the cylinder is 6.5 cm and the total height of the structure is 12.8cm. The volume V of the structure lies between:

[CDS-2014-1]

- (a) 370 cm³ and 380 cm³
(b) 380cm³ and 390 cm³
(c) 390 cm³ and 400 cm³
(d) None of these

24. The diameter of the base of a cone is 6 cm and its altitude is 4cm. What is the approximate curved surface area of the cone?

[CDS-2014-1]

- (a) 45cm² (b) 47cm²
(c) 49cm² (d) 51cm²

25. A drainage tile is a cylindrical shell 21 cm long. The inside and outside diameter are 4.5 cm and 5.1 cm, respectively. What is the volume of the clay required for the tile?

[CDS-2014-1]

- (a) $6.96\pi\text{cm}^3$ (b) $6.76\pi\text{cm}^3$
(c) $5.76\pi\text{cm}^3$ (d) None of these

26. For a plot of land of 100m × 80m, the level to be raised by spreading the earth from stack of a rectangular base 10m × 8m with vertical section being a trapezium of height 2m. The top of the stack is 8m × 5m. By how many centimeter can be level raised?

- (a) 3cm (b) 2.5cm
(c) 2 cm (d) 1.5cm
- [CDS-2014-1]**
27. A cube has each edge 2 cm and a cuboid is 1cm long, 2cm wide and 3 cm high. The paint in a certain container is sufficient to paint an area equal to 54cm^2 . Which one of the following is correct?
- [CDS-2014-1]**
- (a) Both cube and cuboid can be painted
(b) Only cube can be painted
(c) Only cuboid can be painted
(d) Neither cube nor cuboid can be painted
28. A cone of radius r cm and height h cm is divided into two parts by drawing a plane through the middle point of its height and parallel to the base. What is the ratio of the volume of the original cone to the volume of the smaller cone?
- [CDS-2014-1]**
- (a) 4:1 (b) 8:1
(c) 2:1 (d) 6:1
29. If 64 identical small spheres are made out of big sphere of diameter 8cm, then what is surface area of each small sphere?
- [CDS-2014-1]**
- (a) πcm^2 (b) $2\pi\text{cm}^2$
(c) $4\pi\text{cm}^2$ (d) $8\pi\text{cm}^2$
30. The dimensions of a field are 15m by 12m. A pit 8m long, 2.5 m wide and 2m deep is dug in one corner of the field and the earth removed is evenly spread over the remaining area of the field. The level of the field is raised by:
- [CDS-2014-1]**
- (a) 15cm (b) 20 cm
(c) 25cm (d) $\frac{200}{9}\text{cm}$
31. What is the diameter of the largest circle lying on the surface of a sphere of surface area 616 sq cm ?
- [CDS-2014-1]**
- (a) 14cm (b) 10.5cm
(c) 7cm (d) 3.5cm
32. The volume of a hollow cube is $216x^3$. What surface area of the largest sphere which be enclosed in it?
- [CDS-2014-1]**
- (a) $18\pi x^2$ (b) $27\pi x^2$
(c) $36\pi x^2$ (d) $72\pi x^2$
33. A cylinder circumscribes a sphere. What is the ratio of volume of the sphere to that of the cylinder?
- [CDS-2014-2]**
- (a) 2:3 (b) 1:2
(c) 3:4 (d) 3:2
- For the next three (03) items that follow:**
A right triangle having hypotenuse 25cm and legs in the ratio 3 : 4 is made to revolve about its hypotenuse ($\pi = 3.14$)
34. What is the volume of the double cone so formed?
- [CDS-2014-2]**
- (a) 3124 cm^3 (b) 3424 cm^3
(c) 3768 cm^3 (d) 3924 cm^3
35. What is the surface area of the double cone so formed?
- [CDS-2014-2]**
- (a) 1101.2cm^2 (b) 1111.4cm^2
(c) 1310.4 cm^2 (d) 1318.8 cm^2
36. Consider the following statements:
1.The volume of the cone generated when the triangle is made to revolve about its longer leg is same as the volume of the cone generated when the triangle is made to revolve about its shorter leg.
2.The sum of volume of the cone generated when the triangle is made to revolve about its longer leg and the volume of the cone generated when the triangle is made to revolve about its shorter leg is equal to the volume of the double cone generated when the triangle is made to revolve about its hypotenuse?
Which of the above statements is / are correct?
- [CDS-2014-2]**
- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
37. Consider the following statements in respect of four spheres A, B, C and D having respective radii 6, 8, 10 and 12cm.
1.The surface area of sphere C is equal to the sum of surface area of sphere A and B.
2.The volume of sphere D is equal to the sum of volumes of sphere A, B and C.
Which of the above statements is/are correct?
- [CDS-2014-2]**
- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

38. The diameter of a metallic sphere is 6cm. The sphere is melted and drawn into a wire of uniform circular cross section. If the length of the wire is 36m, then what is its radius equal to?

[CDS-2014-2]

- (a) 0.1cm (b) 0.01cm
(c) 0.0001cm (d) 1.0cm

39. The area of the three adjacent faces of a cuboidal box are x , $4x$ and $9x$ sq unit. What is the volume of the box?

[CDS 2014(2)]

- (a) $6x^2$ cu unit (b) $6x^{3/2}$ cu unit
(c) $3x^{3/2}$ cu unit (d) $2x^{3/2}$ cu unit

40. What is the number of pairs of perpendicular planes in a cuboids?

[CDS 2014(2)]

- (a) 4 (b) 8
(c) 12 (d) None of these

41. How many equilateral triangles can be formed by joining any three vertices of a cube?

[CDS 2014(2)]

- (a) 0 (b) 7
(c) 8 (d) None of these

42. What is the maximum distance between two points of a cube of side 2 cm?

[CDS 2014(2)]

- (a) $\sqrt{3}$ (b) $2\sqrt{3}$
(c) $4\sqrt{3}$ (d) $2\sqrt{2}$

For the next two (02) items that follow:

A toy is in the form of a cone mounted on the hemisphere with the same radius. The diameter of the base of the conical portion is 12cm and its height is 8cm.

43. What is the total surface area of the toy?

[CDS 2014(2)]

- (a) 132π cm² (b) 112π cm²
(c) 96π cm² (d) 66π cm²

44. What is the volume of the toy?

[CDS 2014(2)]

- (a) 180π cm³ (b) 240π cm³
(c) 300π cm³ (d) 320π cm³

45. A cylindrical vessel of radius 4cm contains water. A solid sphere of radius 3cm is lowered into the water until it is completely immersed. The water level in the vessel will rise by:

[CDS-2015-1]

- (a) 1.5cm (b) 2cm
(c) 2.25cm (d) 4.5cm

46. If the radius of a sphere is increased by 10%, then the volume will be increased by:

[CDS-2015-1]

- (a) 33.1% (b) 30%
(c) 50% (d) 10%

47. A sphere and a cube have same surface area. The ratio of square of their volumes is:

[CDS-2015-1]

- (a) $6:\pi$ (b) $5:\pi$
(c) 3:5 (d) 1:1

48. The radius of a sphere is equal to the radius of the base of a right circular cone, and the volume of the sphere is double the volume of the cone. The ratio of the height of the cone to the radius of its base is:

[CDS-2015-1]

- (a) 2:1 (b) 1:2
(c) 2:3 (d) 3:2

49. Water flows through a cylindrical pipe of internal diameter 7cm at the rate of 5m/s. The time, in minutes, the pipe would take to fill an empty rectangular tank $4m \times 3m \times 2.31m$ is:

[CDS-2015-1]

- (a) 28 (b) 24
(c) 20 (d) 12

50. The total outer surface area of a right circular cone of height 24cm with a hemisphere of radius 7 cm upon its base is:

[CDS-2015-1]

- (a) 327π square cm (b) 307π square cm
(c) 293π square cm (d) 273π square cm

51. A rectangular block of wood having dimensions $3m \times 2m \times 1.75$ has to be painted on all its faces. The layer of paint must be 0.1 mm thick. Paint comes in cubical boxes having their edges equal to 10cm. The minimum number of boxes of paint to be purchased is:

[CDS-2015-1]

- (a) 5 (b) 4
(c) 3 (d) 2

52. The diagonals of three faces of a cuboid are 13, $\sqrt{281}$ and 20 linear units. Then the total surface area of the cuboid is:

[CDS-2015-1]

- (a) 650 square units (b) 658 square units
(c) 664 square units (d) 672 square units

53. A rectangular paper of 44cm long and 6 cm wide is rolled to form a cylinder of height equal to width of the paper. The radius of the base of the cylinder so rolled is:

[CDS-2015-1]

- (a) 3.5cm (b) 5cm
(c) 7cm (d) 14cm

54. If three metallic sphere of radii 6cm, 8cm and 10 cm are melted to form a single sphere, then the diameter of the new sphere will be:

[CDS-2015-1]

- (a) 12cm (b) 24cm
(c) 30cm (d) 36cm

55. If the height of a right circular cone is increased by 200% and the radius of the base is reduced by 50% then the volume of the cone:

[CDS-2015-1]

- (a) Remains unaltered
(b) Decreases by 25%
(c) Increases by 25%
(d) Increases by 50%

56. A pipe with square cross-section is supplying water to a cistern which was initially empty. The area of cross section is 4cm² and the nozzle velocity of water is 40 m/s. The dimensions of the cistern are 10m×8m×6m. Then the cistern will be full in:

[CDS-2015-2]

- (a) 9.5hours (b) 9hours
(c) 8hours 20 min (d) 8 hours

57. A hollow cylindrical drum has internal diameter of 30cm and a height of 1m. What is the maximum number of cylindrical boxes of diameter 10 cm and height 10cm each that can be packed in the drum?

[CDS-2015-2]

- (a) 60 (b) 70
(c) 80 (d) 90

58. Consider the following statements:

- 1.If the height of a cylinder is doubled, the area of the curved surface is doubled
- 2.If the radius, of a hemispherical solid is doubled, its total surface area becomes fourfold?

Which of the above statements is/are correct?

[CDS-2015-2]

- (a) 1Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

59. A large water tank has the shape of a cube. If 128 m³ of water is pumped out, the water level goes down by 2m. Then the maximum capacity of the tank is:

[CDS-2015-2]

- (a) 512m³ (b) 480m³
(c) 324m³ (d) 256m³

60. From the solid gold in the form of a cube of side length 1cm, spherical solid balls each having the surface area $\pi^{1/3}$ cm² are to be made. Assuming that there is no loss of the material in the process of making the balls, the maximum number of balls made will be:

[CDS-2015-2]

- (a) 3 (b) 4
(c) 6 (d) 9

61. Water is filled in a container in such a manner that its volume doubles every 5 minutes. If it takes 30 minutes for the container to be full, in how much time will it be one fourth full?

[CDS-2015-2]

- (a) 7.5 minutes (b) 15 minutes
(c) 20 minutes (d) 17.5 minutes

62. 30 metallic cylinders of same size are melted and cast in the form of cones having the same radius and height as those of the cylinders.

Consider the following statements:

Statement I: A maximum of 90 cones will be obtained

Statement II: The curved surface of the cylinder can be flattened in the shape of a rectangle but the curved surface of the cone when flattened has the shape of triangle.

Which one of the following is correct in respect of the above?

[CDS-2015-2]

- (a)Both statement I and Statement II are correct and Statement II is the correct explanation of Statement I
(b)Both statement I and Statement II are correct and Statement II is not the correct explanation of Statement I.
(c)Statement I is correct but Statement II is not correct
(d)Statement I is not correct but Statement II is correct

63. A water tank, open at the top, is hemispherical at the bottom and cylindrical

above it. The radius is 12m and the capacity is $3312\pi\text{m}^3$. The ratio of the surface area of the spherical and cylindrical portions is:

[CDS-2015-2]

- (a) 3:5 (b) 4:5
(c) 1:1 (d) 6:5

64. The areas of three mutually perpendicular faces of a cuboid are x , y , z . If V is the volume, then xyz is equal to:

[CDS-2015-2]

- (a) V (v) V^2
(c) $2V$ (d) $2V^2$

65. Let V be the volume of an inverted cone with vertex at origin and the axis of the cone is along positive y -axis. The cone is filled with water up to half of its height. The volume of water is:

[CDS-2015-2]

- (a) $\frac{V}{8}$ (b) $\frac{V}{6}$
(c) $\frac{V}{3}$ (d) $\frac{V}{2}$

66. If the surface area of a cube is 13254cm^2 , then the length of its diagonal is:

[CDS-2015-2]

- (a) $44\sqrt{2}\text{ cm}$ (b) $44\sqrt{3}\text{ cm}$
(c) $47\sqrt{2}\text{ cm}$ (d) $47\sqrt{3}\text{ cm}$

67. How many spherical bullets each of 4 cm in diameter can be made out of a cube of lead whose edge is 44cm?

[CDS-2015-2]

- (a) 2541 (b) 2551
(c) 2561 (d) 2571

68. A river 2.5 m deep and 45m wide is flowing at the speed of 3.6km/hour. The amount of water that runs into the sea per minutes is:

[CDS-2015-2]

- (a) 6650m^3 (b) 6750m^3
(c) 6850m^3 (d) 6950m^3

69. Three rectangles R_1 , R_2 and R_3 have the same area. Their lengths x_1 , x_2 and x_3 respectively are such that $x_1 < x_2 < x_3$. If V_1 , V_2 and V_3 are the volumes of the cylinders formed from the rectangles R_1 , R_2 and R_3 respectively by joining the parallel sides along the breadth, then which one of the following is correct?

[CDS 2015-2]

- (a) $V_3 < V_2 < V_1$ (b) $V_1 < V_3 < V_2$
(c) $V_1 < V_2 < V_3$ (d) $V_3 < V_1 < V_2$

70. How many right angled triangles can be formed by joining the vertices of a cuboid?

[CDS 2015-2]

- (a) 24
(b) 28
(c) 32
(d) None of the above

71. The area of four walls of a room is 120m^2 . The length of the room is twice its breadth. If the height of the room is 4 m, what is area of the floor?

[CDS-2016-2]

- (a) 40m^2 (b) 50m^2
(c) 60m^2 (d) 80m^2

72. The ratio of the curved surface area to the total surface area of a right circular cylinder is 1:2. If the total surface area is 616cm^2 , what is the volume of the cylinder?

[CDS-2016-2]

- (a) 539cm^3 (b) 616cm^3
(c) 1078cm^3 (d) 1232cm^3

73. A cubic meter of copper weighing 9000kg is rolled into a square bar 9 m long. An exact cube is cut off from the bar. How much does the cube weigh?

[CDS-2016-2]

- (a) 1000kg (b) $\frac{1000}{3}\text{ kg}$
(c) 300kg (d) $\frac{500}{3}\text{ kg}$

74. Into a conical tent of radius 8.4m and vertical height 3.5m, how many full bags of wheat can be emptied, if space required for the wheat in each bag is 1.96m^3 ?

[CDS-2016-2]

- (a) 264 (b) 201
(c) 132 (d) 105

75. A building is the form of a cylinder surmounted by a hemispherical dome on the diameter of the cylinder. The height of the building is three times the radius of the base of the cylinder. The building contains $67\frac{1}{21}\text{m}^3$ of air. What is the height of the building?

[CDS-2016-2]

- (a) 6m (b) 4m
(c) 3m (d) 2m

76. The radius of the base and the height of a solid right circular cylinder are in the ratio 2:3 and its volume is 1617 cm^3 . What is the total surface area of the cylinder?

[CDS-2016-2]

- (a) 462cm^2 (b) 616cm^2
(c) 770cm^2 (d) 786cm^2

77. The cost of painting a spherical vessel of diameter 14cm is Rs.8008. What is the cost of painting per square centimeter?

[CDS-2016-2]

- (a) Rs.8 (b) Rs.9
(c) Rs.13 (d) Rs.14

78. A drinking glass of height 24cm is in the shape of frustum of a cone and its diameters of its bottom and top circular ends are 4cm and 18 cm respectively. If we take capacity of the glass as $\pi x \text{ cm}^3$, then what is the value of x?

[CDS-2016-2]

- (a) 824 (b) 1236
(c) 1628 (d) 2472

79. Rain water from a roof $22\text{m} \times 20\text{m}$ drains into a cylindrical vessel having diameter of base 2m and height 3.5m. If the vessel is just full, what is the rainfall?

[CDS-2016-2]

- (a) 3.5cm (b) 3cm
(c) 2.5cm (d) 2cm

80. The height of a cone is 60cm. A small cone is cut of at the top by a plane parallel to the base and its volume is $\frac{1}{64}$ the volume of original cone. What is the height from the base at which the section is made?

[CDS-2016-2]

- (a) 15cm (b) 20cm
(c) 30cm (d) 45cm

81. What is the volume of a sphere of radius 3cm?

[CDS-2016-2]

- (a) $36\pi \text{ cm}^3$ (b) $18\pi \text{ cm}^3$
(c) $9\pi \text{ cm}^3$ (d) $6\pi \text{ cm}^3$

Directions (for next three): A tent of a circus is made of canvas and is in the form of right circular cylinder and right circular cone above it. The height and diameter of the cylindrical part of the tent are 5 m and 126m respectively. The total height of the tent is 21 m.

82. What is the slant height of the cone?

[CDS-2016-2]

- (a) 60m (b) 65m
(c) 68m (d) 70m

83. What is the curved surface area of the cylinder?

[CDS-2016-2]

- (a) 1980m^2 (b) 2010m^2
(c) 2100m^2 (d) 2240m^2

84. How many square metres of canvas are used?

[CDS-2016-2]

- (a) 14450 (b) 14480
(c) 14580 (d) 14850

85. If the radius of a right circular cone is increased by p% without increasing its height, then what is the percentage increase in the volume of the cone?

[CDS-2017-1]

- (a) p^2 (b) $2p^2$
(c) $\frac{p^2}{100}$ (d) $p\left(2 + \frac{p}{100}\right)$

86. If the surface area of a sphere is reduced to one-ninth of the area, its radius reduces to:

[CDS-2017-1]

- (a) One-fourth (b) One-third
(c) One fifth (d) One ninth

87. Ice-cream, completely filled in a cylinder of diameter 35cm and height 32cm, is to be served by completely filling identical disposable cones of diameter 4cm and height 7 cm. The maximum number of persons that can be served in this way is:

[CDS-2017-1]

- (a) 950 (b) 1000
(c) 1050 (d) 1100

88. A ball of radius 1 cm is put into a cylindrical pipe so that it fits inside the pipe. If the length of the pipe is 14m, what is the surface area of the pipe?

[CDS-2017-1]

- (a) 2200 square cm
(b) 4400 square cm
(c) 8800 square cm
(d) 17600 square cm

89. A cylinder of height 2x is circumscribed by a sphere of radius 2x such that the circular ends of the cylinder are two small circles on the sphere. What is the ratio of the curved

surface area of the cylinder to the surface area of the sphere?

[CDS-2017-2]

- (a) $\sqrt{3}:4$ (b) $\sqrt{3}:3$
(c) $\sqrt{3}:2$ (d) $\sqrt{3}:1$

90. A cylindrical vessel 60cm in diameter is partially filled with water. A sphere 30cm in diameter is gently dropped into the vessel and is completely immersed. To what further height will the water in the cylinder rise?

[CDS-2017-2]

- (a) 20cm (b) 15cm
(c) 10cm (d) 5cm

91. The vertical angle of a right circular cone is $\frac{\pi}{2}$ and the slant height is $\sqrt{2}r$ cm. What is the volume of the cone in cubic cm?

[CDS-2017-2]

- (a) πr^3 (b) $9\pi r^3$
(c) $\frac{\pi r^3}{3}$ (d) $3\pi r^3$

92. The radii of the frustum of a right circular cone are in the ratio 2:1. What is the ratio of the volume of the frustum of the cone to that of the whole cone?

[CDS-2017-2]

- (a) 1:8 (b) 1:4
(c) 3:4 (d) 7:8

93. From a solid cylinder whose height is 8cm and of base radius 6cm, a conical cavity of height 8cm and a base radius 6cm is formed by hollowing out. What is the inner surface area of the cavity?

[CDS-2017-2]

- (a) 6π square cm (b) 8π square cm
(c) 10π square cm (d) 60π square cm

94. A tent has been constructed which is in the form of a right circular cylinder surmounted by a right circular cone whose axis coincides with the axis of the cylinder. If the radius of the base of the cylinder is 50m, the height of the cylinder is 10m and the total height of the tent is 15m, then what is the capacity of the tent in cubic metres?

[CDS-2017-2]

- (a) 37500π (b) $\frac{87500\pi}{3}$

(c) $\frac{26500\pi}{3}$

(d) 25000π

95. Two rectangular sheet of sizes $2\pi \times 4\pi$ and $\pi \times 5\pi$ are available. A hollow right circular cylinder can be formed by joining a pair of parallel sides of any sheet. What is the maximum possible volume of the circular cylinder that can be formed this way?

[CDS-2017-2]

- (a) $4\pi^2$ (b) $8\pi^2$
(c) $1.25\pi^2$ (d) $6.25\pi^2$

96. A hollow cube is formed by joining six identical squares. A rectangular cello tape of length 4 cm and breadth 0.5 cm is used for joining each pair of edges. What is the total area of cello tape used?

[CDS 2017-2]

- (a) 12 square cm (b) 24 square cm
(c) 36 square cm (d) 48 square cm

97. The radius and slant height of a right circular cone are 5 cm and 13 cm respectively. What is the volume of the cone?

[CDS-2018-1]

- (a) $100\pi\text{cm}^3$ (b) $50\pi\text{cm}^3$
(c) $65\pi\text{cm}^3$ (d) $169\pi\text{cm}^3$

98. If the ratio of the radius of the base of a right circular cone to its slant height is 1:3, what is the ratio of the total surface area to the curved surface area?

[CDS-2018-1]

- (a) 5:3 (b) 3:1
(c) 4:1 (d) 4:3

99. A right circular cone is sliced into a smaller cone and a frustum of a cone by a plane perpendicular to its axis. The volume of the smaller cone and the frustum of the cone are in the ratio 64:61. Then their curved surface areas are in the ratio:

[CDS-2018-1]

- (a) 4:1 (b) 16:9
(c) 64:61 (d) 81:64

100. In a room whose floor is a square of side 10m, an equilateral triangular table of side 2 m is placed. Four book shelves of size 4m $\times$ 1m $\times$ 9m are also placed in the room. If half of the rest of the area in the room is to be carpeted at the rate of Rs100 per square metre, what is the cost of carpeting (approximately)?

[CDS-2018-1]

- (a) Rs7600 (b) Rs5635
(c) Rs4113 (d) Rs3200

101. What is the volume of a cone of maximum volume cut out from a cube of edge $2a$ such that their bases are on the same plane?

[CDS-2018-1]

- (a) πa^2 (b) $\frac{\pi a^3}{3}$
(c) $\frac{2\pi a^3}{3}$ (d) $\frac{3\pi a^3}{4}$

102. There are as many square centimeter in the surface area of a sphere as there are cubic centimeter in its volume. What is the radius of the sphere?

[CDS-2018-1]

- (a) 4cm (b) 3cm
(c) 2cm (d) 1cm

103. The curved surface area of right circular cone is 1.76m^2 and its base diameter is 140cm. What is the height of the cone?

[CDS-2018-1]

- (a) 10cm (b) $10\sqrt{2}$ cm
(c) $20\sqrt{2}$ cm (d) $10\sqrt{15}$ cm

104. A cube of maximum volume (each corner touching the surface from inside) is cut from a sphere. What is the ratio of the volume of the cube to that of the sphere?

[CDS-2018-1]

- (a) $3:4\pi$ (b) $\sqrt{3}:2\pi$
(c) $2:\sqrt{3}\pi$ (d) $4:3\pi$

105. If the ratio of the circumference of the base of a right circular cone of radius r to its height is 3:1, then what is the area of the curved surface of the cone?

[CDS-2018-1]

- (a) $3\pi r^2$ (b) $\frac{2\pi r^2 \sqrt{4\pi^2 + 9}}{3}$
(c) $\frac{\pi r^2 \sqrt{\pi^2 + 1}}{3}$ (d) $\frac{\pi r^2 \sqrt{4\pi^2 + 9}}{3}$

106. The surface area of closed cylindrical box is 352 square cm. If its length is 10cm, then what is its diameter? $\left(\text{Use } \pi = \frac{22}{7}\right)$

[CDS-2018-1]

- (a) 4cm (b) 8cm
(c) 9.12cm (d) 19.26cm

107. Walls (excluding roof and floors) of 5 identical rooms having length, breadth and height 6m, 4m, and 2.5m respectively are to be painted. Out of five rooms two rooms have one square window each having a side 2.5m. Paints are available only in cans of 1 litre; and 1 litre of paint can be used for painting 20 square metres. The number of cans required for painting is:

[CDS-2018-1 & 2]

- (a) 10 (b) 12
(c) 13 (d) 14

108. The lateral surface area of a cone is 462cm^2 . Its slant height is 35cm. The radius of the base of the cone is:

[CDS-2018-2]

- (a) 8.4cm (b) 6.5cm
(c) 4.2cm (d) 3.2cm

109. A semi circular plate is rolled up to form a conical surface. The angle between the generator and the axis of the cone is:

[CDS-2018-2]

- (a) 60° (b) 45°
(c) 30° (d) 15°

110. A solid right cylinder is of height π cm. If its lateral surface area is half its total surface area, then the radius of its base is:

[CDS-2018-2]

- (a) $\pi/2$ (b) πcm
(c) $1/\pi\text{cm}$ (d) $2/\pi\text{cm}$

111. A rectangular block of length 20 cm, breadth 15 cm and height 10 cm is cut up into exact number of equal cubes. The least possible number of cubes will be:

[CDS-2018-2]

- (a) 12 (b) 16
(c) 20 (d) 24

112. If the diagonal of a cube is of length l , then the total surface area of the cube is:

[CDS-2018-2]

- (a) $3l^2$ (b) $\sqrt{3}l^2$
(c) $\sqrt{2}l^2$ (d) $2l^2$

113. Two cones have their heights in the ratio 1:3. If the radii of their bases are in the ratio 3:1, then the ratio of their volumes will be:

[CDS-2018-2]

- (a) 1:1 (b) 2:1
(c) 3:1 (d) 9:1

Direction for (next two):

A cube is inscribed in a sphere. A right circular cylinder is within the cube touching all the vertical faces. A right circular cone is inside the cylinder. Their heights are same and the diameter of the cone is equal to that of cylinder.

- 114.** What is the ratio of the volume of the sphere to that of the cone?

[CDS-2019-1]

- (a) $6\sqrt{3}:1$ (b) $7:2$
(c) $3\sqrt{3}:1$ (d) $5\sqrt{3}:1$

- 115.** What is the ratio of the volume of the cube to that of the cylinder?

[CDS-2019-1]

- (a) 4:3 (b) 21:16
(c) 14:11 (d) 45:32

- 116.** Consider the following statements:

1. The surface area of the sphere is $\sqrt{5}$ times the curved surface area of the cone.
 2. The surface area of the cube is equal to the curved surface area of the cylinder.
- Which of the above statements is/are correct?

[CDS-2019-1]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Direction for (next two):

The sum of length, breadth and height of a cuboid is 22 cm and the length of its diagonal is 14cm.

- 117.** What is the surface area of the cuboid?

[CDS-2019-1]

- (a) 288cm^2
(b) 216cm^2
(c) 144cm^2
(d) Cannot be determined due to insufficient data

- 118.** If S is the sum of the cubes of the dimensions of the cuboid and V is its volume, then what is $(S-3V)$ equal to?

[CDS-2019-1]

- (a) 572cm^3
(b) 728cm^3
(c) 1144cm^3
(d) None of the above

- 119.** The volume of a spherical balloon is increase by 700%. What is the percentage increase in its surface area?

[CDS-2019-1]

- (a) 300% (b) 400%
(c) 450% (d) 500%

- 120.** A hollow right circular cylindrical vessel of volume V whose diameter is equal to its height, is completely filled with water. A heavy sphere of maximum possible volume is then completely immersed in the vessel. What volume of water remains in the vessel?

[CDS-2019-2]

- (a) $\frac{V}{2}$ (b) $\frac{V}{3}$
(c) $\frac{2V}{3}$ (d) $\frac{V}{4}$

- 121.** Two cylinder of equal volume have their heights in the ratio 2:3. What is the ratio of their radii?

[CDS-2019-2]

- (a) $\sqrt{3}:1$ (b) $\sqrt{3}:\sqrt{2}$
(c) $2:\sqrt{3}$ (d) $\sqrt{3}:2$

- 122.** A right circular cylinder has a diameter of 20 cm and its curved surface area is 1000cm^2 . What is the volume of the cylinder?

[CDS-2019-2]

- (a) 4000cm^3 (b) 4500cm^3
(c) 5000cm^3 (d) 5200cm^3

- 123.** A hollow sphere of external and internal diameters 6cm and 4cm respectively is melted into a cone of base diameter 8cm. What is the height of the cone?

[CDS-2019-2]

- (a) 4.75cm (b) 5.50cm
(c) 6.25cm (d) 6.75cm

- 124.** A solid metallic cylinder of height 10 cm and radius 6cm is melted to make two cones in the ratio of volume 1:2 and of the same height as 10cm. What is the percentage increase in the flat surface area?

[CDS-2019-2]

- (a) 25% (b) 50%
(c) 75% (d) 100%

- 125.** Three copper spheres of radii 3cm, 4cm and 5 cm are melted to form a large sphere. What is its radius?

[CDS-2019-2]

- (a) 12cm (b) 10cm
(c) 8cm (d) 6cm

- 126.** The volume of a hemisphere is 155232m^3 . What is the radius of the hemisphere?

[CDS-2019-2]

- (a) 40cm (b) 42cm
(c) 38cm (d) 36cm

- 127.** A bucket is in the form of a truncated cone. The diameters of the base and top of the bucket are 6cm and 12cm respectively. If the height of the bucket is 7cm, what is the capacity of the bucket?

[CDS-2019-2]

- (a) 535cm^3 (b) 462cm^3
(c) 234cm^3 (d) 166cm^3

- 128.** A right circular cone has height 8cm. If the radius of its base is 6cm, then what is its total surface area?

[CDS-2019-2]

- (a) $96\pi\text{cm}^2$ (b) $69\pi\text{cm}^2$
(c) $54\pi\text{cm}^2$ (d) $48\pi\text{cm}^2$

- 129.** Six cubes, each with 12cm edge are joined end to end. What is the surface area of resulting cuboid?

[CDS-2019-2]

- (a) 3000cm^2 (b) 3600cm^2
(c) 3744cm^2 (d) 3777cm^2

- 130.** The area of three adjacent faces of a cuboid are x, y and z. If V is the volume of the cuboid, then which one of the following is correct?

[CDS-2019-2]

- (a) $V=xyz$ (b) $V^2=xyz$
(c) $V^3=xyz$ (d) $V=(xyz)^2$

- 131.** A thin rod of length 24 feet is cut into rods of equal size and joined so as to form a skeleton cube. What is the area of one of the faces of the largest cube thus constructed?

[CDS-2019-2]

- (a) 25 square feet (b) 24 square feet
(c) 9 square feet (d) 4 square feet

- 132.** The thickness of a cylinder is 1 foot, the inner radius of the cylinder is 3feet and height is 7feet. To paint the inner surface it

requires one litre of a particular colour. How much quantity of the same colour is required to paint all the surface of the cylinder?

[CDS-2020-1]

- (a) $\frac{7}{3}$ litre (b) $\frac{3}{2}$ litre
(c) $\frac{8}{3}$ litre (d) $\frac{10}{3}$ litre

- 133.** How many cubic metre of earth is to be dug out to dig a well of radius 1.4m and depth 5m?

[CDS-2020-1]

- (a) 30.2cubic metre (b) 30.4 cubic metre
(c) 30.6cubic metre (d) 30.8 cubic metre

- 134.** Out of 4 identical balls of radius r, 3 balls are placed on a plane such that each ball touches the other two balls. The 4th ball is placed on them such that this ball touches all the three balls. What is the distance of center of 4th ball from the plane?

[CDS-2020-1]

- (a) $2\sqrt{\frac{2}{3}}r$ unit (b) $\frac{\sqrt{3}+2\sqrt{2}}{\sqrt{2}}r$ unit
(c) $\frac{r}{3-2\sqrt{2}}r$ unit (d) $\frac{\sqrt{3}+2\sqrt{r}}{\sqrt{3}}r$ unit

- 135.** A right circular cylinder just encloses a sphere. If p is the surface area of the sphere and q is the curved surface area of the cylinder, then which one of the following is correct?

[CDS-2020-1]

- (a) $p=q$ (b) $p=2q$
(c) $2p=q$ (d) $2p=3q$

- 136.** If H, C and V are respectively the height, curved surface area and volume of a cone, then what is $3\pi VH^3 + 9V^2$ equal to?

[CDS-2020-2]

- (a) C^2H^2 (b) $2C^2H^2$
(c) $5C^2H^2$ (d) $7C^2H^2$

- 137.** How many solid lead balls each of diameter 2mm can be made from a solid lead ball of radius 8cm?

[CDS-2020-2]

- (a) 512 (b) 1024
(c) 256000 (d) 512000

- 138.** The surface areas of two sphere are in the ratio 1:4. What is the ratio of their volumes?

[CDS-2020-2]

- (a) 1:16 (b) 1:12
(c) 1:10 (d) 1:8

139. The length, breadth and height of a brick are 20cm, 15 cm and 10cm respectively. The number of bricks required to construct at wall with dimensions 45m length, 0.15m breadth and 3 m height is:

[CDS-2020-2]

- (a) 12450
(b) 11250
(c) 6750
(d) None of the above

140. The volumes of two cones are in the ratio 1:4 and their diameters are in the ratio 4:5. What is the ratio of their heights?

[CDS-2020-2]

- (a) 25:64 (b) 16:25
(c) 9:16 (d) 5:9

141. If p, q, r, s and t represent length, breadth, height, surface area and volume of a cuboid respectively, then what is $\frac{1}{p} + \frac{1}{q} + \frac{1}{r}$ equal to?

[CDS-2020-2]

- (a) $\frac{s}{t}$ (b) $\frac{2t}{s}$
(c) $\frac{s}{2t}$ (d) $\frac{2s}{t}$

142. The radii of the flat circular faces of a bucket are x and 2x. If the height of the bucket is 3x, what is the capacity of the bucket? (assume $\pi = \frac{22}{7}$).

[CDS-2020-2]

- (a) $11x^3$ (b) $22x^3$
(c) $44x^3$ (d) $55x^3$

143. A river 3m deep and 40 m wide is flowing at the rate of 2 km/h and falls into the sea. What is the amount of water in litres that will fall into the sea from this river in a minute?

[CDS-2020-2]

- (a) 40,00,000 litres (b) 4,00,000 litres
(c) 40,000 litres (d) 4,000 litres

144. A cloth of 3m width is used to make a conical tent 12 m in diameter with a slant height of 7m. What is the length of the cloth? $\left(\text{take } \pi = \frac{22}{7} \right)$

[CDS-2021-1]

- (a) 21m (b) 28m
(c) 44m (d) 66m

145. A sphere of diameter 6cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 6cm. If the sphere is completely submerged in water, then by how much will the surface level of water be raised?

[CDS-2021-1]

- (a) 0.5cm (b) 1cm
(c) 1.5cm (d) 2cm

146. The surface area of a cube is equal to that of sphere. If x is the volume of the cube and y is the volume of the sphere, then what is $x^2 : y^2$ equal to?

[CDS-2021-1]

- (a) $\pi:6$ (b) $6:\pi$
(c) $\pi:3$ (d) $3:\pi$

147. A cone and a hemisphere have equal bases and equal volumes. What is the ratio of the height of the cone to the radius of the hemisphere?

[CDS-2021-1]

- (a) 1:1 (b) 2:1
(c) 3:2 (d) 4:3

148. A solid sphere of diameter 60mm is melted to stretch into a wire of length 144cm. What is the diameter of the wire?

[CDS-2021-1]

- (a) 0.5cm (b) 1cm
(c) 1.5cm (d) 2cm

149. The ratio of the radius of base to the height of a cylinder is 2:3. If the volume of the cylinder is 1617 cm^2 , then what is the curved surface area of the cylinder? $\left(\text{take } \pi = \frac{22}{7} \right)$.

[CDS-2021-1]

- (a) 242cm^2 (b) 385cm^2
(c) 462cm^2 (d) 770cm^2

150. A metal solid cube of edge 24cm is melted and made into three small cubes. If the edges of two small cubes are 12 cm and 16cm, then what is the surface area of the third small cube?

[CDS-2021-1]

- (a) 1200cm^2 (b) 1800cm^2
(c) 2400cm^2 (d) 3600cm^2

- 151.** A conical vessel whose internal radius is 5cm and height 24cm is full of water. The water is emptied into a cylindrical vessel with internal radius 10cm. What is the height to which the water rises?

[CDS-2021-1]

- (a) 1cm (b) 2cm
(c) 3cm (d) 4cm

- 152.** A metal solid cube of side 22 cm is melted to make a cone of height 21 cm. What is the 'radius of the base of the cone?

$\left(\text{take } \pi = \frac{22}{7} \right)$

[CDS-2021-1]

- (a) 11cm (b) 16.5cm
(c) 22cm (d) 27.5cm

- 153.** A cone of height 24 cm has a curved surface area 550 cm². What is the ratio of its radius to slant height? $\left(\text{take } \pi = \frac{22}{7} \right)$

[CDS-2021-1]

- (a) $\frac{5}{12}$ (b) $\frac{5}{13}$
(c) $\frac{7}{25}$ (d) $\frac{7}{27}$

- 154.** A rectangular paper is 44cm long and 22cm wide. Let x be the volume of the largest cylinder formed by rolling the paper along its length and y be the volume of the largest cylinder formed by rolling the paper along its width. What is the ratio of x to y?

$\left(\text{take } \pi = \frac{22}{7} \right)$

[CDS-2021-1]

- (a) 1:1 (b) 2:1
(c) 1:2 (d) 3:2

- 155.** A hollow spherical shell is made up of a metal of density 3g/cm³. If the internal and external radii are 5 cm and 6cm respectively, then what is the mass of the shell?

$\left(\text{take } \pi = \frac{22}{7} \right)$

[CDS-2021-1]

- (a) 1144g (b) 1024g
(c) 840g (d) 570g

- 156.** The difference between the outside and the inside surface area of a cylindrical pipe 14cm long is 44 cm². The pipe is made of 99cm³ of metal. If R is the outer radius and

r is the inner radius of the pipe, then what is (R + r) equal to? (Take $\pi = \frac{22}{7}$)

[CDS 2021 (I)]

- (a) 9cm (b) 7.5cm
(c) 6cm (d) 4.5cm

- 157.** A hemispherical bowl of internal radius 18cm contains a liquid. The liquid is filled in small cylindrical bottles of internal radius 3cm and internal height 4cm. What is the number of bottles used to empty the bowl?

[CDS-2021-2]

- (a) 54 (b) 81
(c) 108 (d) 135

- 158.** A hollow spherical shell is made of a metal of density 7 g/cm³. If its internal and external radii are 3 cm and 6cm respectively, then what is the mass of the shell? (take $\pi = \frac{22}{7}$).

[CDS-2021-2]

- (a) 2772g (b) 3322g
(c) 4433g (d) 5544g

- 159.** A cone of height 16cm and diameter 14cm is mounted on a hemisphere of same diameter. What is the volume of the solid thus formed? (take $\pi = \frac{22}{7}$)

[CDS-2021-2]

- (a) 1540 cm³ (b) 1078 cm³
(c) 1048 cm³ (d) 770 cm³

- 160.** 3 cubes each of volume 343 cm³ are joined end to end. What is the total surface area of the resulting cuboid?

[CDS-2021-2]

- (a) 343 cm² (b) 350 cm²
(c) 686 cm² (d) 700 cm²

- 161.** A cubical block of side 14cm is surmounted by a hemisphere of radius 7cm. What is the total surface area of the solid thus formed? (take $\pi = \frac{22}{7}$)

[CDS-2021-2]

- (a) 1330 cm² (b) 1306 cm²
(c) 1296 cm² (d) 1256 cm²

- 162.** How many silver coins, 3.5cm in diameter and of thickness 4 mm, must be melted to form a cuboid of dimension 21cm × 11 cm × 7 cm? (take $\pi = \frac{22}{7}$)

[CDS-2021-2]

- (a) 420 (b) 210
(c) 200 (d) 168

- 163.** A tub is in the shape of a frustum of a cone. The radii of two circular ends of the tub are 105 cm and 42 cm. If the vertical height of the tub is 16 cm, what is its slant height?

[CDS-2021-2]

- (a) 63.5 cm (b) 65 cm
(c) 73.5 cm (d) 75 cm

- 164.** ABC is a triangle right-angled at B with AB = 8 cm and BC = 6 cm. It is made to revolve about its side BC. What is the approximate total surface area of the cone so formed? (take $\pi = \frac{22}{7}$)

[CDS-2021-2]

- (a) 452 cm² (b) 440 cm²
(c) 432 cm² (d) 420 cm²

- 165.** A solid rod consists of a cylinder of height 20 cm and radius 7 cm. It is surmounted by another solid cylinder of height 10 cm and radius 3.5 cm. If 1 cubic metre of rod weighs 10000 kg, what is the mass of the rod?

(take $\pi = \frac{22}{7}$)

[CDS-2021-2]

- (a) 34.65 kg (b) 31.56 kg
(c) 3.465 kg (d) 3.156 kg

Consider the following for the next three (03) items that follow:

A solid consisting of a right circular cone of radius x and height $2x$ standing on a hemisphere of radius x (take $\pi = \frac{22}{7}$)

- 166.** The volume of the solid is equal to that of a:

[CDS-2021-2]

- (a) Sphere of radius x
(b) Sphere of diameter x
(c) Cylinder of radius x
(d) Cylinder of radius $\sqrt{2}x$

- 167.** What is the approximate total surface area of the solid?

[CDS-2021-2]

- (a) $11.2x^2$ (b) $12.5x^2$
(c) $13.3x^2$ (d) $15.1x^2$

- 168.** The solid is placed upright in a right circular cylinder full of water such that it touches the bottom. If the internal radius of cylinder is x and height is $3x$, what is the approximate volume of water left in the cylinder?

[CDS-2021-2]

- (a) $5.04x^3$ (b) $5.09x^3$
(c) $5.14x^3$ (d) $5.24x^3$

- 169.** The length and breadth of a room are 21 m and 16 m respectively. If the length of the longest rod that can be placed in the room is 29 m, then what is the height of the room?

[CDS 2021-2]

- (a) 10 m (b) 11 m
(c) 12 m (d) 13 m

- 170.** Water is trickling out of a completely filled cylindrical tank of height 1 m and diameter 2 m. Every second a spherical droplet 2 cm. Every seconds a spherical droplet of 1 cm radius trickles down from the bottom of the tank. The tank will be emptied in approximately.

[CDS 2022-1]

- (a) 280 hours (b) 260 hours
(c) 230 hours (d) 210 hours

- 171.** The length, breadth and height of a cuboid are in the ratio 27 : 8 : 1. The cuboid is melted and recast into a cube. If p is the surface area of the cuboid and q is the surface area of the cube, then what is p/q equal to?

[CDS 2022-1]

- (a) $\frac{247}{108}$ (b) $\frac{251}{108}$
(c) $\frac{503}{216}$ (d) $\frac{505}{216}$

- 172.** In a shower, 5 cm of rain falls. What is the volume of water that falls on 2 hectare area of land?

[CDS 2022-1]

- (a) 100 cubic meter
(b) 1000 cubic meter
(c) 4000 cubic meter
(d) 10000 cubic meter

- 173.** The radius and height of a right circular cone are in the ratio 3 : 7. If the volume of the cone is 528 cm³, then what is the height of the cone? (Take $\pi = \frac{22}{7}$)

[CDS 2022-1]

- (a) 3.5 cm (b) 7.0 cm
(c) 10.5 cm (d) 14.0 cm

- 174.** A cylindrical pipe has inner diameter of 14 cm. Water flows through it at a rate of

154 litres per minute. What is the speed of water in km/hr? (Take $\pi = \frac{22}{7}$)

[CDS 2022-1]

- (a) 0.5 (b) 0.6
(c) 0.8 (d) 1

175. A spherical ball of lead 6cm in diameter is melted and recast into three spherical balls. The diameters of the balls are 2cm, 4cm and x cm. Which one of the following is correct?

[CDS 2022-1]

- (a) $5\text{cm} < x\text{cm} < 5.2\text{cm}$
(b) $5.2\text{cm} < x\text{cm} < 5.4\text{cm}$
(c) $5.4\text{cm} < x\text{cm} < 5.6\text{cm}$
(d) $5.6\text{cm} < x\text{cm} < 5.8\text{cm}$

176. A rectangular sheet is of length x and breadth y. If p is the volume of the cylinder formed by rolling the sheet along its breadth and q is the volume of the cylinder formed by rolling the sheet along its length, and $q = 2p$, then which one of the following is correct?

[CDS 2022-1]

- (a) $x = y$ (b) $2x = 3y$
(c) $x = 2y$ (d) $3x = 4y$

177. A tall cylindrical container with circular base of radius 18cm contains a good quantity of water. Metal balls each of radius 0.9cm are immersed in it. How many balls are required to raise the water level by 3cm?

[CDS 2022-1]

- (a) 100 (b) 500
(c) 1000 (d) 1500

178. The surface area of a cube of length x is equal to the surface area of a sphere of radius y. Consider the following statements:

1. $2x > 3y$
2. The volume of the cube is greater than the volume of the sphere.

Which of the above statements is/are correct?

[CDS 2022-1]

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

179. A closed box is built of wood of uniform thickness. Its external dimensions are

12cm, 10cm and 8cm. If the inner surface area is 376 square cm, then what is the thickness of the wood?

[CDS 2022 (II)]

- (a) 0.5cm (b) 1cm
(c) 1.5cm (d) 2cm

180. The height of a cone is 30cm. A small cone is cut off at the top by a plane parallel to its base. If its volume is $\frac{1}{27}$ of the volume of the given cone, then what is the height of the frustum of the cone?

[CDS 2022 (II)]

- (a) 10cm (b) 12cm
(c) 18cm (d) 20cm

181. A rectangular metal sheet is of length 24 cm and breadth 18cm. From each of its corners a square of side x cm is cut off and an open box is made of the remaining sheet. If the volume of the box is 640 cubic cm, then what is the value of x?

[CDS 2022 (II)]

- (a) 2 (b) 3
(c) 4 (d) 6

182. Let x, y, z be the length, breadth, height of a cuboid. If its volume is 400 cubic cm and total surface area is 340 square cm, then what is $x^{-1} + y^{-1} + z^{-1}$ equal to?

[CDS 2022 (II)]

- (a) $\frac{17}{20}$ (b) $\frac{17}{40}$
(c) $\frac{7}{10}$ (d) $\frac{9}{10}$

183. The length of a room is $\frac{21}{16}$ times its breadth and breadth is $\frac{4}{7}$ times its height. If H is the height of the room and L is the longest rod that can be placed in the room, then which one of the following is correct?

[CDS 2022 (II)]

- (a) $12L = 29H$ (b) $9L = 25H$
(c) $7L = 23H$ (d) $8L = 13H$

Consider the following for the next three (03) items the follow:

A conical vessel of radius 12cm and height 16cm is filled with water. A sphere is lowered into water and its size is such that

it touches the sides of the vessel and it is just immersed.

184. What is the radius of the sphere?

[CDS 2022 (II)]

- (a) 5cm (b) 6cm
(c) 6.5cm (d) 7cm

185. How much water will remain in the vessel after the overflow?

[CDS 2022 (II)]

- (a) $288\pi\text{mL}$ (b) $360\pi\text{mL}$
(c) $480\pi\text{mL}$ (d) $500\pi\text{mL}$

186. What is the ratio of lateral surface area of the vessel to the surface area of the sphere?

[CDS 2022 (II)]

- (a) $\frac{4}{3}$ (b) $\frac{3}{2}$
(c) $\frac{5}{3}$ (d) 2

187. The diameter of a sphere made of copper is 3cm. The sphere is melted and recast into a wire. If the length of the wire is half metre, then what is the diameter of the wire?

[CDS 2022 (II)]

- (a) 0.3cm (b) .045cm
(c) 0.6cm (d) 0.75cm

188. The volume of a cuboid is 3600 cubic cm. The area of two adjacent faces are 225 square cm and 144 square cm. What is the area of the other adjacent face?

[CDS – 2023 (1)]

- (a) 400 square cm (b) 360 square cm
(c) 320 square cm (d) 300 square cm

189. Two times the total surface area of a solid right circular cylinder is three times its curved surface area. If h is the height and r is the radius of the base of the cylinder, then which one of the following is correct?

[CDS – 2023 (1)]

- (a) $h=r$ (b) $h=2r$
(c) $2h=3r$ (d) $3h=4r$

190. A solid iron ball is melted and 64 smaller solid balls of equal size are made using the entire volume of iron. What is the ratio of the surface area of the larger ball to the sum of the surface area of all the smaller balls?

[CDS – 2023 (1)]

- (a) 0.25 (b) 0.5
(c) 0.75 (d) 1

191. The surface area of a cube is increased by 25%. If p is the percentage increase in its length, then which one of the following is correct?

[CDS – 2023 (1)]

- (a) $16 < p < 18$ (b) $14 < p < 16$
(c) $12 < p < 14$ (d) $10 < p < 12$

192. A solid cube is cut into two cuboids of equal volume. What is the ratio of total surface area of the given cube to that of one the cuboids?

[CDS – 2023 (1)]

- (a) 2:1 (b) 3:2
(c) 4:3 (d) 5:3

193. The length of a diagonal of a cuboid is 11 cm. The surface area is 240 square cm. What is the sum of its length, breadth and height?

[CDS – 2023 (1)]

- (a) 16cm (b) 17cm
(c) 18cm (d) 19cm

194. Three solid led spheres of diameters 6cm, 8cm and 10cm are melted together and recast as a solid sphere. What is the percentage diminution of the surface area as compared to the sum of the surface area of the three sphere?

[CDS-2023 (2)]

- (a) 25% (b) 26%
(c) 27% (d) 28%

195. A solid sphere of radius 3cm is melted to form a hollow cylinder of height 4cm and external diameter 10cm. What is the thickness of the cylinder?

[CDS-2023 (2)]

- (a) 0.42 cm (b) 0.46cm
(c) 0.5cm (d) 1.00cm

196. A sphere of radius 5cm is dropped in a right circular cylindrical vessel partly filled with water. The radius of the cylindrical vessel is 10cm. If the sphere is completely submerged in water, by how much will the level of water, rise in the cylindrical vessel?

[CDS-2023 (2)]

- (a) $\frac{5}{3}\text{cm}$ (b) $\frac{5}{2}\text{cm}$
(c) 1cm (d) $\frac{5}{6}\text{cm}$

197. What is the radius of the sphere passing through the corners of the cuboid with edges 8cm, 12cm and 24cm?

[CDS-2023 (2)]

- (a) 10.5cm (b) 14cm
(c) 21cm (d) 28cm

198. A lamp shade is in the shape of a part of a cone and its top and bottom ends are circles whose circumferences are respectively 30cm and 40cm. The perpendicular distance between the ends is 6 cm. If the cone were to be completed, then how far would its vertex be from the top end?

[CDS-2023 (2)]

- (a) 20cm (b) 18cm
(c) 12cm (d) 9cm

Consider the following for the next two (02) items that follow:

A right conical cap just covers two spheres placed one above the other on a table such that it touches both the spheres. Let r be the radius of the smaller sphere and R be the radius of the bigger sphere. Let 2θ be the vertical angle of the cone.

199. What is the height of the cone?

[CDS-2023 (2)]

- (a) $\frac{2r^2}{R-r}$ (b) $\frac{2R^2}{R-r}$
(c) $\frac{2(r^2+R^2)}{R-r}$ (d) $\frac{r^2+R^2}{R-r}$

200. What is the radius of the base of the cone?

[CDS-2023 (2)]

- (a) $\frac{2r^2 \tan \theta}{R-r}$ (b) $\frac{2R^2 \tan \theta}{R-r}$
(c) $\frac{2(r^2+R^2) \tan \theta}{R-r}$ (d) $\frac{(r^2+R^2) \tan \theta}{R-r}$

201. The length, breadth and height of a cuboid are increased by 10%, 20% and 50% respectively. What is the percentage increase in volume of the cuboid?

[CDS-2024 (1)]

- (a) 100% (b) 99%
(c) 98% (d) 50%

202. A cube whose edge is 14 cm long has on each of its faces a circle of 7 cm radius painted yellow. What is the total area of unpainted surface? (Take $\pi = \frac{22}{7}$)

[CDS-2024 (1)]

- (a) 126 square cm (b) 189 square cm
(c) 252 square cm (d) 315 square cm

203. From a circular metal plate of radius 7 cm and thickness 0.16 mm, a sector is cut off containing an angle 150° . The remaining piece is molded into a spherical bead of radius r . What is the value of r in cm?

[CDS-2024 (1)]

- (a) 0.35 (b) 0.7
(c) 1.05 (d) 1.4

ANSWER KEY

1.	d	2.	b	3.	a	4.	b	5.	d	6.	b	7.	a	8.	d	9.	a	10.	a
11.	c	12.	c	13.	b	14.	a	15.	d	16.	a	17.	c	18.	b	19.	c	20.	d
21.	c	22.	a	23.	a	24.	b	25.	d	26.	d	27.	a	28.	b	29.	c	30.	c
31.	a	32.	c	33.	d	34.	c	35.	d	36.	d	37.	c	38.	a	39.	b	40.	b
41.	d	42.	b	43.	a	44.	b	45.	c	46.	a	47.	a	48.	a	49.	b	50.	d
51.	c	52.	c	53.	c	54.	a	55.	b	56.	c	57.	d	58.	c	59.	a	60.	c
61.	c	62.	c	63.	b	64.	b	65.	a	66.	c	67.	a	68.	b	69.	c	70.	d
71.	b	72.	c	73.	b	74.	c	75.	a	76.	c	77.	c	78.	a	79.	c	80.	d
81.	a	82.	b	83.	a	84.	d	85.	d	86.	b	87.	c	88.	c	89.	a	90.	d
91.	c	92.	d	93.	d	94.	b	95.	b	96.	b	97.	a	98.	d	99.	b	100.	c
101.	c	102.	b	103.	d	104.	c	105.	d	106.	b	107.	b	108.	c	109.	c	110.	b
111.	d	112.	d	113.	c	114.	a	115.	c	116.	d	117.	a	118.	c	119.	a	120.	b
121.	b	122.	c	123.	a	124.	b	125.	d	126.	b	127.	b	128.	a	129.	c	130.	b
131.	d	132.	c	133.	d	134.	d	135.	a	136.	a	137.	d	138.	d	139.	c	140.	a
141.	c	142.	b	143.	d	144.	c	145.	b	146.	a	147.	b	148.	b	149.	c	150.	c
151.	b	152.	c	153.	c	154.	c	155.	a	156.	d	157.	c	158.	d	159.	a	160.	c
161.	a	162.	a	163.	b	164.	a	165.	a	166.	a	167.	c	168.	d	169.	c	170.	d
171.	b	172.	b	173.	d	174.	b	175.	b	176.	c	177.	c	178.	c	179.	b	180.	d
181.	c	182.	b	183.	a	184.	b	185.	c	186.	c	187.	c	188.	a	189.	b	190.	a
191.	b	192.	b	193.	d	194.	d	195.	d	196.	a	197.	b	198.	a	199.	b	200.	b
201.	c	202.	c	203.	b														

CAPF PYQ

1. A solid spherical ball made of iron is melted and two new balls are made whose diameters are in the ratio of 1:2. The ratio of the volume of the smaller new ball to the original ball is:

- (a) 1:3
(c) 2:9

- [CAPF 2022]
(b) 1:5
(d) 1:9

ANSWER KEY

1.	d																		
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